

BO-112 Plus Pembrolizumab for Patients With Anti-PD-1-Resistant Advanced Melanoma: Phase II Clinical Trial SPOTLIGHT-203

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ABSTRACT

PURPOSE Patients with anti-PD-1-resistant melanoma (MEL) have no well-defined standard of care. BO-112 is a synthetic, double-stranded RNA (poly I:C) nanoplexed with polyethylenimine that when administered intratumorally has showed in patients with solid tumors potential to revert this resistance. We report efficacy and safety of the phase II clinical trial of intratumoral BO-112 plus intravenous pembrolizumab for patients with anti-PD-1-resistant MEL (ClinicalTrials.gov identifier: [NCT04570332](https://clinicaltrials.gov/ct2/show/study?term=NCT04570332)).

METHODS Forty-two patients were treated with intratumoral BO-112 once every week for 7 weeks and then once every 3 weeks (up to 2 mg and up to eight lesions per treatment) combined with 200 mg pembrolizumab once every 3 weeks until progressive disease, unacceptable toxicity, death, or up to 1 year. Primary end point was RECIST 1.1 objective response rate (ORR) by independent central radiology review in modified intention-to-treat population (mITT; patients evaluable for response) with 20% ORR positivity threshold. Secondary key end points were progression-free survival (PFS), overall survival (OS), duration of response (DOR), and safety.

RESULTS For mITT, there were 40 patients and the ORR was 25%, with 10% complete, 15% partial, and 40% stable disease, with nonachieved (NA) median DOR (95% CI, 8.3 to NA). For ITT, there were 42 patients, and the median PFS and OS were 3.7 months (95% CI, 2.2 to 9.2) and NA (95% CI, 9.9 to NA), respectively, with 54% patients alive at 24 months. The combination was well tolerated: 16 patients (38.1%) experiencing $\geq$ G3-4 adverse events, four (9.5%) drug-related, and no deaths related to treatment.

CONCLUSION The clinical trial has met its primary end point (ORR) making BO-112 with pembrolizumab a potential strategy to revert anti-PD-1 resistance in patients with MEL. PFS results are in line with other clinical trials in anti-PD-1-resistant scenario, with promising OS data.

ACCOMPANYING CONTENT

- [Data Sharing Statement](#)
- [Data Supplement](#)
- [Protocol](#)

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INTRODUCTION

Anti-PD-1 antibodies nivolumab and pembrolizumab, whether as monotherapy^{1,2} or in combination with ipilimumab² or relatlimab,³ have become the standard of care for patients with advanced melanoma (MEL). This benefit has been translated to the adjuvant setting, with a significant prolongation of relapse-free survival in patients with high-risk MEL.⁴⁻⁶

However, up to 70% of the patients will eventually experience progressive disease (PD) during first-line treatment, because of primary and secondary resistance mechanisms.^{2,7}

There is no current standard of care for second line after anti-PD-1-based therapies for patients with advanced MEL or for patients developing PD while on adjuvant treatment or shortly after it. Different strategies have been reported, including antiangiogenics (lenvatinib plus pembrolizumab),

CONTEXT

Key Objective

Can intratumoral BO-112 and intravenous pembrolizumab revert anti-PD-1–resistance in patients with advanced melanoma (MEL)?

Knowledge Generated

This combination demonstrated a 25% objective response rate in patients with MEL resistant to anti-PD-1, with a median progression-free survival of 3.7 months and median overall survival not reached. Fifty-four percent patients were alive at 24 months, with a manageable toxicity profile.

Relevance (G. McArthur)

Overcoming resistance to therapy with anti-PD-1 in MEL is a high priority. This study demonstrates clinical meaningful activity of intralesional synthetic double-stranded RNA combined with pembrolizumab, including in noninjected lesions, paving the way for larger randomized trials.*

*Relevance section written by JCO Associate Editor Grant McArthur, PhD, MBBS.

tumor-infiltrating lymphocytes (TILs), or ipilimumab plus anti-PD-1, with objective response rates (ORRs) around 20%–30% and median progression-free survival (PFS) around 4–5 months.^{8–10} However, the use of ipilimumab in monotherapy and the continuation of anti-PD-1 in monotherapy have demonstrated <10% to 15% of responses.^{11,12}

The connection between innate and adaptive immune system is key to initiate and sustain efficient immune antitumoral responses. The inability of innate immunity (mainly dendritic cells) to create this connection could make the tumor invisible to immune system (cold tumor). Thus, exposing the tumor microenvironment with microorganisms or substances designed to resemble them¹³ could start an innate immune system response that facilitates the recognition of adaptive immunity, making a cold tumor turn into a hot one. Among these strategies, toll-like receptors (TLRs) agonists are currently in an extensive research development.¹³

BO-112 is a synthetic, double-stranded (ds) RNA (poly I:C) nanoplexed with the polyethylenimine, which resembles a dsRNA virus. Intratumoral injection in animal models activates TLR3, MEL differentiation-associated protein 5, and retinoic acid-inducible gene I, overcoming anti-PD-1 resistance, including in models lacking an effective interferon pathway.^{14,15} In a previous phase I clinical trial, intratumoral BO-112 showed a favorable safety profile both as monotherapy or in combination with systemic anti-PD-1 and demonstrated that it could revert anti-PD-1 resistance in patients with solid tumors, including responses in two of 10 patients with MEL.¹⁶ In paired biopsies, BO-112 demonstrated CD4 and CD8 infiltration increase, tumor cell apoptosis, and T-cell activation gene expression increase in patients with clinical benefit, implying a potential immunologic remodeling of cold tumors to hot.¹⁶ The current phase II clinical trial of BO-112 in combination with

pembrolizumab was designed for evaluating its clinical activity in this specific setting.

METHODS

Study Design and Participants

This was a multicenter, open-label phase II clinical trial, conducted in 15 centers from Spain and France (ClinicalTrials.gov identifier: [NCT04570332](https://clinicaltrials.gov/ct2/show/study?term=NCT04570332)). The main inclusion criteria were patients age 18 years and older with Eastern Cooperative Oncology Group 0 or 1, advanced and/or metastatic unresectable stage III or stage IV MEL (cutaneous, acral or mucosal), with known *BRAF* mutational status, that has confirmed PD-1 on anti-PD-1-containing treatment, either as monotherapy, or in combination with other immunotherapies or drugs.

Confirmed PD to anti-PD-1 according to RECIST 1.1 was defined as follows: (1) received ≥ 6 weeks of standard dosing of an approved anti-PD-1 antibody; (2) PD confirmed by a second assessment after 4 weeks, no rapid clinical progression; and (3) PD disease documented within 12 weeks from the last dose of anti-PD-1.

Anti-PD-1–based therapy must have been the last therapy before inclusion, including (1) those who progressed on first-line treatment for advanced disease; (2) those who received adjuvant treatment without progression followed by progression on first-line treatment for advanced disease; and (3) those who received only adjuvant therapy and progressed while on that therapy. *BRAF* inhibitors were not allowed.

Patients should have had at least one tumoral lesion that is measurable according to RECIST 1.1 and amenable for intratumoral injection.

Key exclusion criteria were uveal MEL; previous grade 3–4 immune-related adverse events (AEs) requiring systemic steroids for >2 weeks; previous intratumoral treatments or contraindications for them; any anticancer therapy within 4 weeks, except for anti-PD-1–based therapy and palliative radiotherapy; clinically active brain metastases; and/or carcinomatous meningitis.

This trial was approved by central ethics committee in Spain (Hospital General Universitario Vall d'Hebron, Barcelona) and France (Centre Léon Bérard, Lyon), by health authorities in both countries, and by each participating center, and complied with Good Clinical Practice Guidelines, the Declaration of Helsinki, and local laws. All patients provided written, informed consent before any protocol-related procedure or treatment.

Procedures

Administration of BO-112 was performed once every week for the first 7 weeks and then once every 3 weeks in combination with 200 mg flat dose pembrolizumab once every 3 weeks. BO-112, if needed, was administered guided by ultrasound or occasionally computed tomography (CT) scan guidance, including visceral lesions if deemed safe by collaboration with interventional radiologists. The maximum dose per session of BO-112 was 2 mg (3.4 mL) and the minimum dose at cycle 1 (C1) was 1 mg. It was allowed to administer smaller doses in following cycles in the case of response resulting in a single injectable lesion smaller than 15 mm. The injected dose was divided in up to eight different lesions. Volume and dose administered per lesion were adjusted to the longest diameter of a given lesion (Data Supplement, Table S1, online only). The treatment was continued until PD, unacceptable toxicity, death, patients' withdrawal, or up to 2 years. Administration of BO-112 may be interrupted independently of pembrolizumab treatment in the following situations: previously injected lesion(s) has decreased to a size for which it is not feasible to inject or other reason (ie, density). Pembrolizumab dose reduction was not permitted. Injectable lesions could be target or nontarget lesions. Patients who discontinued therapy were followed until death, up to 24 months after first dose, or conclusion of the study.

Efficacy was assessed by imaging (CT scan and/or MRI as well as digital photographs), beginning 8 weeks from C1, followed by week 16 and every 12 weeks until PD (or more frequently if clinically indicated). Response evaluation was performed by independent radiology central review (IRCR; Quibim, Valencia, Spain).

Baseline tumoral archival biopsies were obtained. If feasible, repeated biopsies were performed on treatment at C2 before BO-112 from an injected lesion (if possible, a nontarget lesion) and an optional biopsy between C3 and the end of C4, to

be analyzed by central pathology reviewer, including PD-L1 expression using MEL score⁴⁷ and next-generation sequencing (Pangaea Oncology, Barcelona, Spain).

The AEs were collected from the first dose of the study drug(s) until 90 days after last administration and classified according to National Cancer Institute Common Terminology Criteria for Adverse Events v5.0.

Outcomes

The primary end point was ORR according to RECIST 1.1 criteria evaluated by IRCR in the modified intention-to-treat population (mITT), which consisted of all enrolled patients who had at least one dose of treatment and a baseline and at least one postbaseline tumor response assessment.

For patients with skin disease only, pathologic response was considered, and, in the event of suspected complete response (CR), any residual cutaneous pigmented areas or other residual masses were documented as not containing tumor by a representative biopsy. The Data Monitoring Committee evaluated patients with a best response per investigator by reviewing photographs of all visible lesions and biopsy results.

The key secondary end points included disease control rate (DCR) defined as the sum of CR, partial response (PR), and stable disease (SD) by IRCR and by investigator; duration of response (DOR) by IRCR and by investigator; PFS by IRCR and investigator; overall survival (OS); and safety and tolerability. Exploratory analysis included biomarkers from baseline and sequential biopsies.

Statistical Analysis

The sample size was calculated using the Khan one-stage design for smaller trials. H_0 was ORR = 10% (p₀) and was tested against H_1 of ORR >10% at one-sided 5% significance level. Assuming an effect size of ORR = 25% (p_A), 40 participants provided a power of at least 80%. The minimal statistically significant ORR was 20% with at least eight of 40 patients achieving response. Primary end point analysis was performed in mITT, whereas secondary end point analysis was performed in the ITT population. The values of 95% CI were calculated using a Wald estimate and exact (Clopper-Pearson) confidence limits for the binomial proportion.

The survival as well as the median and 95% CI time to event are estimated by means of the Kaplan-Meier method.

Safety-related analyses were performed in the safety population.

All date calculations that are required are made taking date of first dose of study drug date as a reference.

Role of the Funding Source

Highlight Therapeutics S.L. was the sponsor of the study. It was designed by the senior academic authors and representatives of Highlight Therapeutics S.L. and Merck & Co, Inc, Rahway, NJ. Data were collected by clinical investigators, and analyzed and interpreted in collaboration with the sponsor. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

RESULTS

Between January 18, 2021, and August 24, 2021, 54 patients were enrolled and 12 excluded for not fulfilling eligibility criteria. Forty-two received at least one dose of BO-112 and pembrolizumab (ITT) and 40 were evaluable for the primary end point (mITT; Data Supplement, Table S2). Baseline patient characteristics are presented in Table 1.

At the time of study entry, 65% of patients (27/42) had PD, 23.8% had SD (10/42), and 11.9% (5/42) had achieved either PR or CR as their best response to prior anti-PD-1–based therapy; 16.7% (7/42) had skin disease only with cutaneous and/or subcutaneous lesions, 19% (8/42) had soft tissue and/or lymph node involvement, and the remaining 64.3% (27/42) had visceral metastatic disease. The mean time from last dose of anti-PD-1 to first dose of BO-112 and pembrolizumab was 11.5 weeks (95% CI, 6.8 to 16.1). The treatment exposure to BO-112 and pembrolizumab are described in the Data Supplement (Table S3).

The ORR according to RECIST 1.1 evaluated by IRCR in the mITT population was 25% (95% CI, 12.7 to 41.2). All responses were confirmed at a second evaluation by IRCR. Overall, 10% were CR (4/40), 15% PR (6/40), and 42.5% SD (17/40), meaning a DCR of 67.5% (Fig 1). Additionally, three of the 16 patients with RECIST v1.1 SD had complete pathologic response in at least one sequential biopsy.

With a median follow-up of 15.5 months (95% CI), the DOR evaluated by IRCR was 85.7% at 9 and 12 months (95% CI, 33.4 to 97.9), with the median DOR not reached, with only two events (Fig 2).

In a post hoc analysis, best overall response (BOR) was analyzed descriptively according to key prognostic and basal factors (responders v nonresponders). Of note, patients with acral MEL and/or lactate dehydrogenase (LDH) $>3\times$ upper limit of normal had no benefit in terms of response. On the contrary, two of the three patients with mucosal MEL enrolled achieved a PR and one SD. Table 2 summarizes the BOR according to key subgroups in the mITT.

Responses were observed in both injected and noninjected target lesions, including distant visceral metastases. Up to 53% of patients had reduction in their injected lesion and 25% of patients had reduction also in noninjected lesions

TABLE 1. Patients' Baseline Characteristics

Characteristic	Total ITT (n = 42)
Sex, No. (%)	
Male	24 (57)
Female	18 (43)
Age, years, median (minimum-maximum)	66 (27-88)
Type of MEL, No. (%)	
Acral	9 (21)
Cutaneous	30 (71)
Mucosal	3 (7)
BRAF status, No. (%)	
Mutated	7 (17)
Wild-type	35 (83)
PD-L1 expression, No. (%)	
Negative (MEL score 0 and 1)	11 (26.24)
Positive (MEL score 2-5)	24 (57.1)
Not available/not tested	7 (16.7)
M stage, No. (%)	
Unresectable III	4 (10)
M1a-b	21 (50)
M1c-d	17 (40)
LDH, No. (%)	
Very high ($3\times >ULN$)	2 (5)
High ($>ULN$)	16 (38)
Normal	26 (62)
Previous treatment line, No. (%)	
Ipilimumab-nivolumab	6 (14)
Anti-PD-1 monotherapy	32 (76)
Other anti-PD-1 combinations	4 (10)
Previous treatment indication, No. (%)	
Adjuvant	10 (24)
Advanced disease	32 (76)
Duration of previous treatment line, weeks, median (minimum-maximum)	23 (3-128)
Type of previous resistance according to SITC, ⁷ No. (%)	
Primary	26 (62)
Secondary	16 (38)
Tumor burden SOD target, mm, mean (95% CI)	89 (67 to 110)

NOTE. BRAF status was determined as per local standards and confirmed centrally. PD-L1 was exploratory and determined by 22C3 immunohistochemistry using MEL score, being positive if ≥ 2 .¹⁷ PD during adjuvant treatment was considered primary resistance. Abbreviations: ITT, intention-to-treat; LDH, lactate dehydrogenase; MEL, melanoma; PD, progressive disease; SITC, Society for Immunotherapy of Cancer; SOD, sum of diameters; ULN, upper limit of normal.

(Data Supplement, Fig S1). Five patients had all their target lesions injected.

With a median follow-up of 15.5 months (95% CI), the median PFS in the ITT population was 3.7 months (95% CI, 2.6 to 9.2), with 44.4% and 32.3% patients being progression-free at 6 months and 12 months, respectively (Fig 3A). In a post hoc analysis, when excluding the

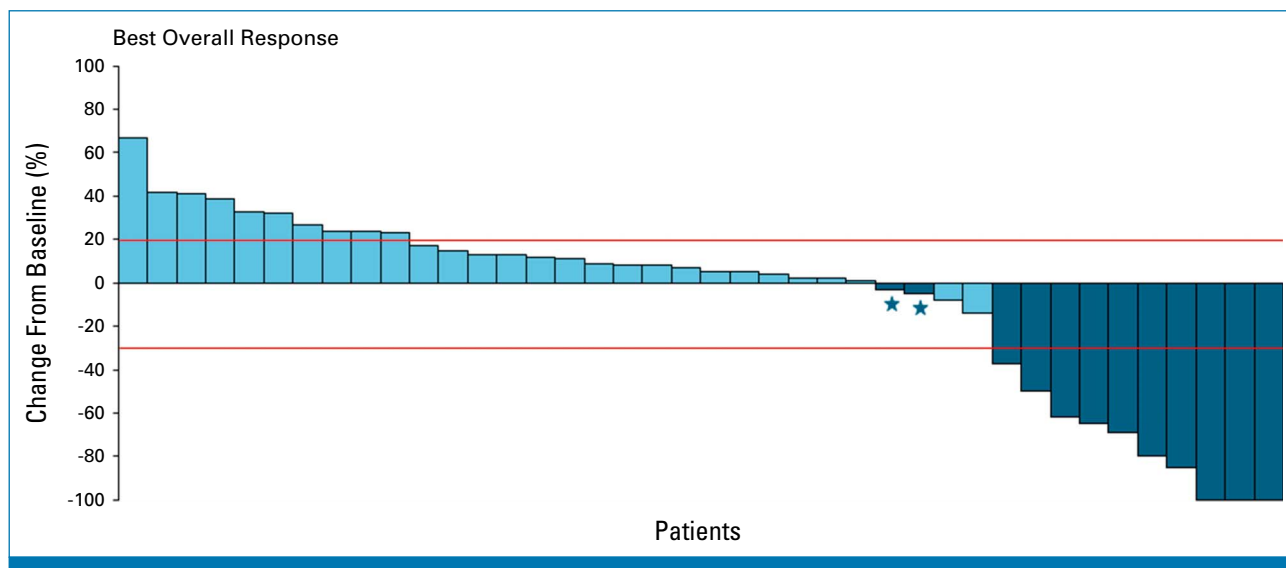


FIG 1. Waterfall plot of best overall responses by IRCR in mITT evaluated by IRCR. Two patients with skin disease only (stars), despite having a RECIST SD, achieved a pCR as evaluated by DMC. DMC, Data Monitoring Committee; IRCR, independent radiology central review; mITT, modified intention-to-treat population; pCR, pathologic complete response; SD, stable disease.

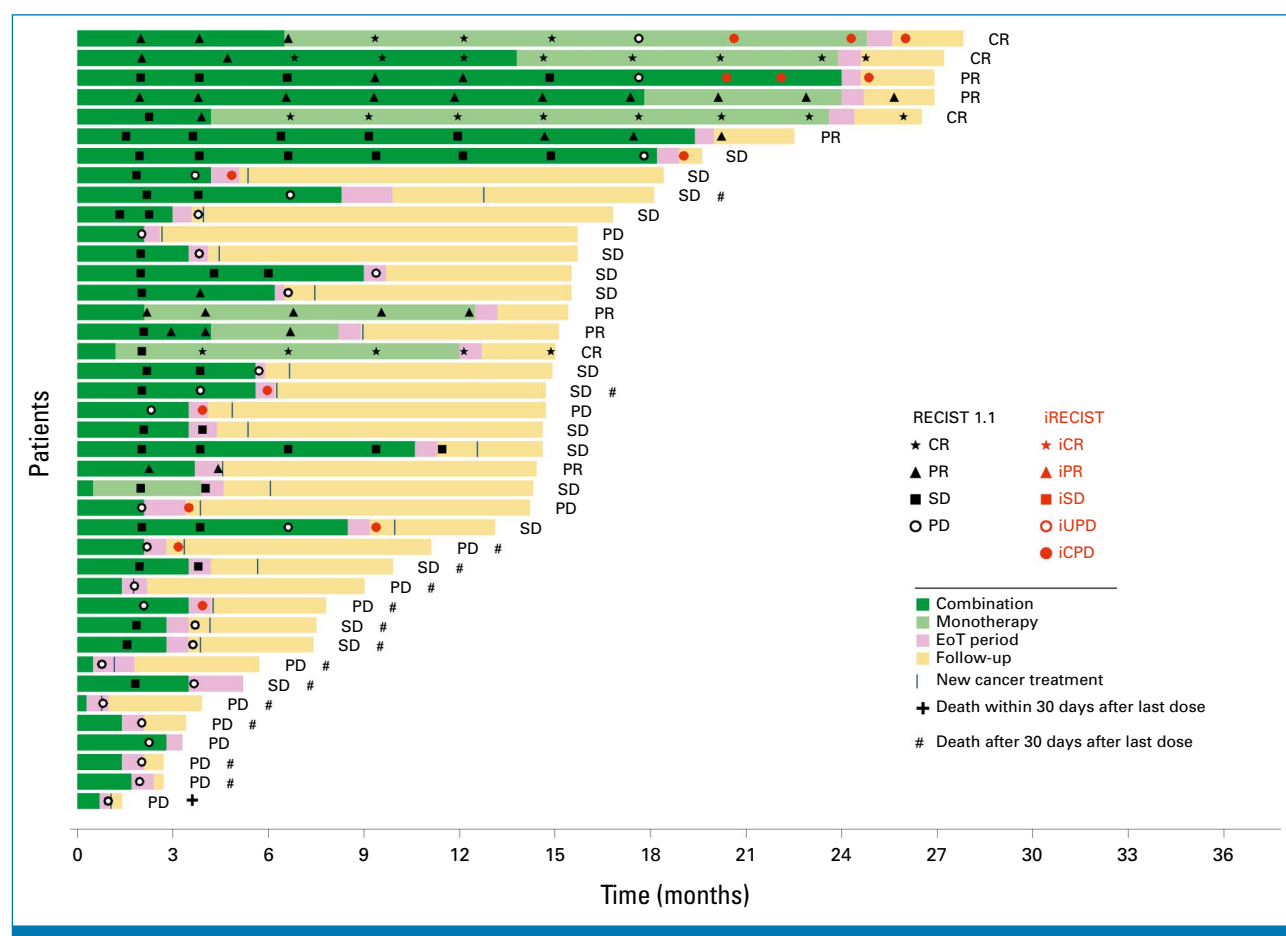


FIG 2. Swimmer plot for duration of treatment and response time-points according to IRCR response evaluation. CR, complete response; iCPD, immune confirmed progressive disease; iCR, immune complete response; iPR, immune partial response; iSD, immune stable disease; IRCR, independent radiology central review; iRECIST, immune RECIST; iUPD, immune unconfirmed progressive disease; PD, progressive disease; PR, partial response; SD, stable disease.

TABLE 2. Univariate Post Hoc Analysis of BOR According to Key Patient Baseline Characteristics in the mITT Population

Total mITT (n = 40)	No.	OR	P
Sex (male v female)	22 v 18	0.444	.2713
Type of melanoma (acral v nonacral)	8 v 32	<0.001	.0221
<i>BRAF</i> status (mutated v native)	7 v 33	6.000	.0421
PD-L1 expression (negative v positive)	11 v 24	0.380	.405
LDH high v normal	15 v 25	0.643	.5674
LDH $\geq 3 \times$ ULN v $< 3 \times$ ULN	2 v 38	<0.001	.2763
Previous treatment line anti-PD-1–based ($\pm$ other agents) v ipilimumab + nivolumab	30 v 10	1.800	.5958
Type of previous resistance according to SITC ⁷ primary v secondary	24 v 16	0.510	.3601

NOTE. PD during adjuvant treatment was considered primary resistance.

Abbreviations: BOR, best overall response; LDH, lactate dehydrogenase; mITT, modified intention-to-treat population; OR, odds ratio; PD, progressive disease; SITC, Society for Immunotherapy of Cancer; ULN, upper limit of normal.

patients with acral subtype and/or LDH $> 3 \times$ UNL, the median PFS was 6.5 months (95% CI, 3.7 to 17.6) with 58.8% and 46.2% progression-free at 6 and 12 months, respectively (Fig 3B).

With a follow-up for OS of up to 24 months, median OS has not been reached (95% CI, 9.9 to nonachieved), with 78.3%, 63.6%, and 53.7% alive at 6, 12, and 24 months, respectively, in the ITT population (Fig 3C).

The safety profile was consistent with that published in the phase I clinical trial.¹⁶

AEs of any cause and regardless of attribution to treatment occurred in 100% of patients (Table 3). Sixteen (38.1%) patients had grade 3–5 AEs, with four (9.5%) considered drug-related. Seven patients (16.7%) died due to an AE (mostly disease progression), which was not related to treatment. Treatment-related AEs occurred in 88.1% (37/42) patients for BO-112 and pembrolizumab, with 9.5% (4/42) being grade 3 or 4. Discontinuation of any trial treatments because of AEs occurred in 16.7% (7/42) patients.

Thirty-five and 24 patients had baseline and paired baseline/on-treatment biopsies, respectively. *NRAS* or *BRAF* mutations were associated with clinical benefit ($P = .02$), which was more evident for patients without *MYC* amplifications ($P = .002$; Data Supplement, Figs S2 and S3). However, baseline PD-L1 expression and basal and post-treatment CD8 expression in paired biopsies did not correlate with benefit ($P = .48$ and $.4$, respectively; Data Supplement, Fig S4).

DISCUSSION

With an ORR of 25%, the primary end point of the SPOT-LIGHT-203 clinical trial has been met, with a significant

proportion of patients having long-lasting responses. This observation is supported by the confirmation in all cases after a second evaluation and by an independent central radiology review. In addition, these responses were observed in patients with a previous confirmed PD, ruling out a possible postprogression response with the maintenance of anti-PD-1, a phenomenon described in $< 10\%$ of patients.¹²

The response rate observed in our study is in line with other phase II clinical trials in anti-PD-1 resistance. Response rate varies from 21.4% for pembrolizumab and lenvatinib in the LEAP004 to 36.4% for TILs in the lifileucel (LN-144) studies.^{8,18} Importantly, these two studies allowed more than one previous line of treatment in contrast to ours, which allowed only one.

The median PFS in the ITT population described in our study is roughly 4 months, meaning that most patients cannot have a meaningful clinical benefit in the short term. This is also observed in the above-mentioned studies, with 4.2 and 4.1 months for lenvatinib and TILs, respectively.^{8,18} Regarding TILs, the phase III academic clinical trial compared this strategy with ipilimumab in patients with advanced MEL (approximately 90% in the second-line, post-PD-1 setting).⁹ TILs outperform ipilimumab in terms of ORR (48.8 v 21.4%) and PFS (median 7.2 months v 3.1) with OS results still immature. It is noteworthy to highlight that 80% of patients had normal LDH levels, in contrast with 59% in our study.

Post hoc analysis of ipilimumab after anti-PD-1 in the KEYNOTE006 study demonstrated a 17.5% ORR.¹⁹ By contrast, the combination of ipilimumab and anti-PD-1 showed 29% ORR and a median PFS of 5 months in a phase II clinical trial.²⁰ Recently, a randomized study evaluated the efficacy of the combination of ipilimumab and nivolumab versus ipilimumab in this setting,¹⁰ including mainly patients with only a previous anti-PD-1–based therapy line for advanced disease. The ORR and median PFS for ipilimumab and nivolumab were 28% and 3 months, respectively, compared with 9% and 2.7 months, respectively, for ipilimumab alone. Thus, this study is considered positive for its primary end point (PFS) and negative for OS (21.7 v 25.7 months).¹⁰ The toxicity profile of ipilimumab and nivolumab combination with 56% of G3–4 AEs is less favorable than that observed with BO-112 and pembrolizumab.

BO-112 is administered intratumorally, thus indirect comparisons could be more appropriate with other intratumoral strategies. The MASTERKEY randomized study did not demonstrate advantages of oncolytic virus talimogene laherparepvec (T-VEC) when administered in combination with intravenous pembrolizumab versus pembrolizumab and intratumoral placebo.²¹ In the second-line setting, this combination yields an ORR in the anti-PD-1–resistant that varied from 0% to 6.7% in the advanced setting (primary v acquired resistance) versus 40% to 46.7% in the

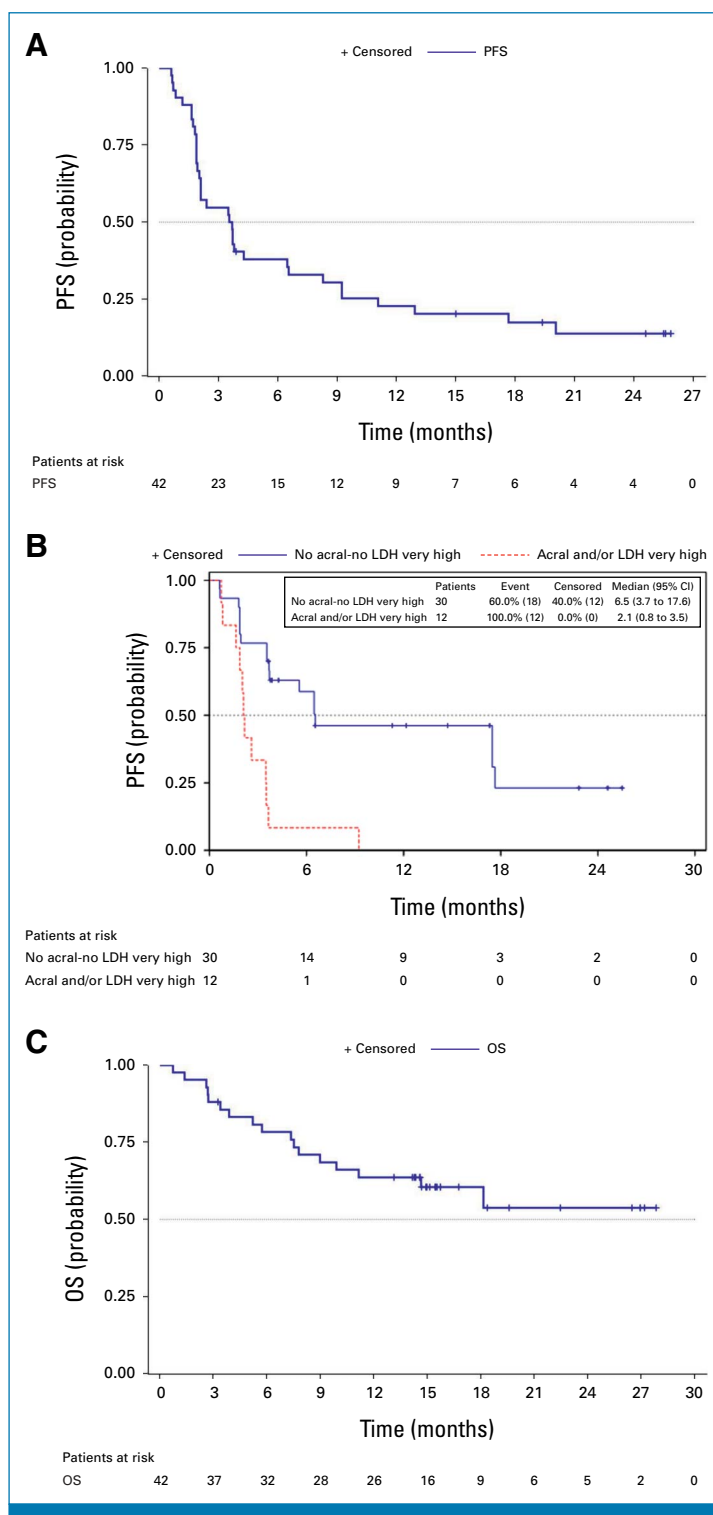


FIG 3. (A) PFS in the ITT population and (B) excluding acral and very high LDH. (C) OS in the ITT population. ITT, intention-to-treat; LDH, lactate dehydrogenase; OS, overall survival; PFS, progression-free survival.

postadjuvant relapse setting (<6 months v > 6 months of disease-free survival).²² In our study, the responses seemed to be better in the advanced setting and for patients with secondary resistance but with differences not as broad as the second-line T-VEC clinical trial. Another oncolytic

modified herpes simplex virus, vusolimogene oderparepvec (RP-1), recently has demonstrated in the same setting as our study similar results in terms of ORR (33.6%) and OS (63.3% at 2 years, median not reached).²³ In contrast to TVEC and RP-1, BO-112 has no biosafety management

TABLE 3. Summary of TEAEs Related to Any Treatment by Worst Grade by Preferred Term (>5%)

Adverse Event Preferred Term	Grade 1, No. (%)	Grade 2, No. (%)	Grade 3, No. (%)	Total, No. (%)
Asthenia	15 (35.7)	11 (26.2)	0	26 (61.9)
Pyrexia	12 (28.6)	7 (16.7)	0	19 (45.2)
Diarrhea	10 (23.8)	3 (7.1)	0	13 (30.9)
Chills	8 (19.1)	4 (9.5)	0	12 (28.6)
Nausea	9 (21.4)	3 (7.1)	0	12 (28.6)
Vomiting	8 (19.1)	3 (7.1)	0	11 (26.2)
Pruritus	7 (16.7)	0	0	7 (16.7)
Headache	4 (9.5)	3 (7.1)	0	7 (16.7)
Influenza-like illness	3 (7.1)	2 (4.8)	0	5 (11.9)
Rash	5 (11.9)	0	0	5 (11.9)
Myalgia	2 (4.8)	0	2 (4.8)	4 (9.5)
Decreased appetite	3 (7.1)	1 (2.4)	0	4 (9.5)
Adrenal insufficiency	0	2 (4.8)	1 (2.4)	3 (7.1)
Arthralgia	2 (4.8)	1 (2.4)	0	3 (7.1)
Fatigue	2 (4.8)	1 (2.4)	0	3 (7.1)
Tremor	2 (4.8)	1 (2.4)	0	3 (7.1)
Injection site pain	2 (4.8)	1 (2.4)	0	3 (7.1)
Pain in extremity	2 (4.8)	1 (2.4)	0	3 (7.1)

Abbreviation: TEAE, treatment emergent adverse event.

issues inherent to oncolytic viruses, making its administration less logistically complicated.

Since BO-112 is a dsRNA, it has among its mechanism of action to be a TLR3 agonist. Thus, the comparison with other TLR agonists in combination is of interest. Tilsotolimod, a TLR9 agonist, demonstrated in a phase I/II clinical trial 22.4% ORR in combination with ipilimumab. In contrast to our study, the immune-related G3–4 AEs were up to 24.2%, because of the known worse safety profile of ipilimumab compared with anti-PD-1 therapies.²⁴ Unfortunately, these results were not reproduced in a later phase III clinical trial comparing intratumoral tilsotolimod and intravenous ipilimumab with ipilimumab, with responses around 8% in both arms and no impact in OS.¹¹

In a phase Ib clinical trial, vidutolimod, another TLR9 agonist, in combination with anti-PD-1 demonstrated a 25% ORR in patients with anti-PD-1–resistant MEL.²⁵ The median PFS was 2.8 months and the G3–4–related AEs for this combination was 45%.²⁵ One important difference with our study is that anti-PD-1 was not mandatory to be the last treatment and more than two thirds of patients received more than two lines of therapy.

The evaluation of response with intratumoral treatments is challenging, since RECIST criteria were designed for systemic treatment but not for local ones. Special criteria (intratumoral-RECIST) are under development to solve this issue, although they should be considered as exploratory, since its validity is still not as solid as RECIST 1.1, which indeed are the most used response criteria in immunotherapy clinical trials.²⁶

In our study, up to eight lesions, whether they were target or nontarget, could be injected per session. In addition, non-injected lesions could be injected during the evolution of the treatment. This makes it difficult to interpret whether the responses were driven by BO-112, pembrolizumab, or both. Indeed, five patients had all their target lesions injected, meaning that they were not evaluable for contribution of these noninjected lesions to the response. This could underestimate a possible abscopal effect of BO-112.

Finally, some of the findings of this clinical trial could orientate to further studies in the randomized setting. Patients with acral MEL had no benefit in terms of ORR. PFS results differ substantially when the patients with acral MEL are excluded in this analysis. This could be due to a poorer response to anti-PD-1 observed in these patients.²⁷ Similarly, very elevated LDH levels could predict poor outcomes. Therefore, similarly to patients with acral MEL, specific and different regimens should be explored. However, the two PRs and one SD response observed in patients with mucosal MEL (another subtype with special poor prognosis) would merit further evaluation. However, for acral, mucosal, and very high LDH, the low population numbers preclude any formal conclusion. In addition, molecular analysis from the pre- and post-BO-112 biopsies could indicate a role of *BRAF*, *NRAS*, and *MYC* alterations in the prediction of clinical benefit. However, further prospective analysis should be done in larger cohorts to confirm this exploratory analysis.

In conclusion, the combination of intratumoral BO-112 and intravenous pembrolizumab has demonstrated that

it could revert the resistance to anti-PD-1-based therapies in 25% of patients with advanced MEL, which at least equals other strategies and with a very favorable toxicity profile. The quality of responses is demonstrated with its duration and the significant percentage of patients alive in the long term, especially for patients who

are not diagnosed with acral MEL and/or very high LDH levels.

Altogether, these findings warrant the exploration of this combination in randomized studies in the anti-PD-1-resistant setting.

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